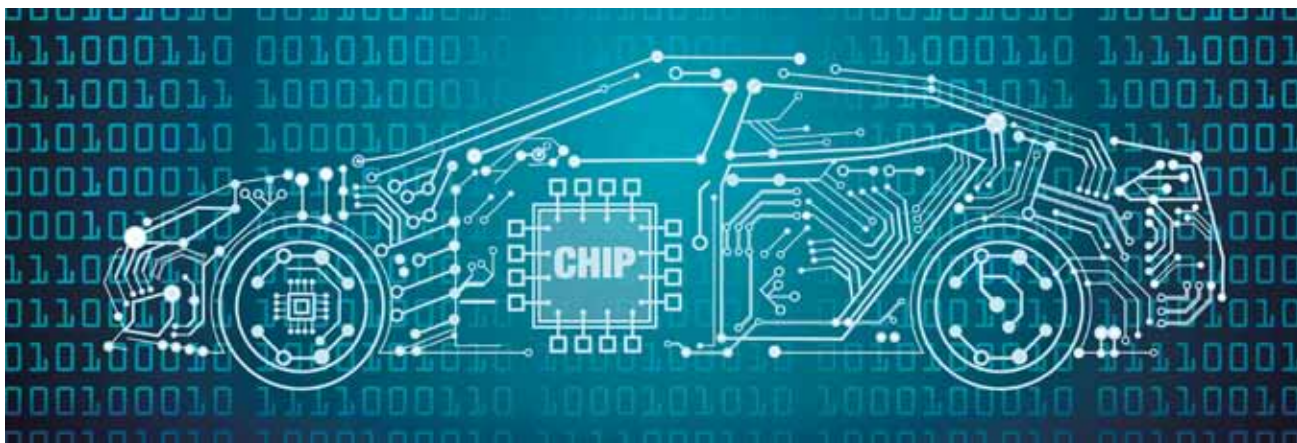


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Special section dedicated to E-Mobility and Vehicle Electronics

What On Earth Is A SOFTWARE DEFINED CAR?

What aspects of your vehicle will be governed by software? How will that impact you? James Falkiner, Technology Analyst at IDTechEx and author of the report 'Connected and Software-Defined Vehicles 2024-2034: Markets, Forecasts, Technologies,' shares his insights on the rapidly evolving landscape of software-defined vehicles and their impact on the automotive industry



JAMES FALKINER is a Technology Analyst at IDTechEx, working in the field of semiconductors with a particular interest in 5G/6G, connected vehicles, and silicon photonic integrated circuits

The concept of a software-defined vehicle (SDV) is emerging as a transformative buzzword in the automotive industry. At its core, an SDV is defined by the significant role software plays in enhancing the user experience. This definition is broad, encompassing a spectrum from basic software like navigation and infotainment systems to advanced AI algorithms powering autonomous vehicles. As vehicles evolve from mere transportation tools to integrated, software-centric experiences, the SDV category becomes increasingly relevant.

In exploring the realm of SDVs, several key aspects come into focus. These

include the level of connectivity, the power and performance of onboard computing, and the extent of vehicle autonomy. Additional factors such as the number and type of displays, the nature of the onboard software, and the vehicle's level of autonomy also play crucial roles in defining a vehicle's software capabilities.

The transition from traditional vehicles to SDVs reflects an evolving automotive landscape, where design and functionality are increasingly intertwined with the user experience. This evolution is marked by a shift from focusing solely on driving performance to enhancing the interactive features such as screens, buttons, and